

Question:

An isosceles right triangle with legs of length \$8\$ is partitioned into \$16\$ congruent triangles as shown. The shaded area is

$\text{(A)}\ 10 \quad \text{(B)}\ 20 \quad \text{(C)}\ 32 \quad \text{(D)}\ 40 \quad \text{(E)}\ 64$

Reasoning + Answer:

<think> To find the shaded area , we need to understand the structure of the isosceles right triangle partitioned into 16 smaller congruent triangles . Each smaller triangle has legs of length (4) , making the area of each small triangle (4) square units . The total number of triangles is (16) , and since the triangles are congruent , we can find the area of the entire triangle by multiplying the area of one small triangle by the total number of triangles . The base and height of the entire isosceles right triangle are both (16) , so its area is $(\frac{1}{2} \times 16 \times 16 = 128)$ square units . Since this area is also the area of 16 smaller triangles , each having an area of (4) square units , we can confirm the total area is indeed (128) square units . Now , the shaded area can be calculated by counting the number of triangles that are shaded and multiplying by the area of one small triangle . From the diagram , we can see that the shaded area consists of 8 smaller triangles . Therefore , the shaded area is $(8 \times 4 = 32)$ square units . </think>

<answer> 32 </answer>

